

CUSTOMER SEGMENTATION ANALYSIS

RFM Analysis • K-Means Clustering • Power BI

Project Report

Submitted as part of the Thiranex Project Assignment

Tools Used

Python • Pandas • NumPy • Scikit-learn • Matplotlib • Seaborn • Power BI

Dataset

Online Retail Dataset (Kaggle)

1. Project Overview

Customer segmentation is the process of dividing customers into groups with similar characteristics or purchasing behavior. This project analyzes online retail transaction data to identify meaningful customer groups and convert those groups into actionable business insights.

The project combines RFM (Recency, Frequency and Monetary) analysis with K-Means clustering. Python was used for data cleaning, customer-level feature engineering and clustering, while Power BI was used to build an interactive dashboard for segment, product and geographic analysis.

2. Project Objectives

- Clean and preprocess the online retail transaction dataset.
- Analyze customer purchasing behavior using Recency, Frequency and Monetary metrics.
- Apply K-Means clustering to identify groups of customers with similar behavior.
- Use silhouette score to select an appropriate number of clusters.
- Analyze purchase patterns and product preferences across customer segments.
- Explore geographic characteristics using customer country.
- Build an interactive Power BI dashboard.
- Generate targeted business recommendations for each customer segment.

3. Dataset Description

The project uses the Online Retail dataset, a transactional dataset containing purchases from an online retail business. Each row represents a product-level transaction.

Column	Description
InvoiceNo	Unique invoice or transaction number.
StockCode	Product identification code.
Description	Description/name of the product.
Quantity	Number of units purchased.
InvoiceDate	Date and time of the transaction.
UnitPrice	Price per unit.
CustomerID	Unique customer identifier.
Country	Customer's country.

The original dataset contained 541,909 transaction records. After preprocessing and removing unsuitable records, 397,884 cleaned transaction records were retained for customer-level analysis.

4. Data Cleaning and Preprocessing

Data preparation was performed in Python using Pandas. The purpose was to remove invalid records and create a reliable transaction table for customer segmentation.

- Removed transactions with missing CustomerID values because customer-level segmentation requires an identified customer.
- Removed cancelled/invalid transactions from the analysis.
- Removed records with non-positive quantities.
- Removed records with non-positive unit prices.
- Converted InvoiceDate into datetime format.
- Calculated transaction revenue as Quantity × UnitPrice.
- Aggregated transaction data to the customer level for RFM analysis.

Stage	Rows / Customers
Original transaction records	541,909
Cleaned transaction records	397,884
Unique customers used for RFM/clustering	4,338

5. RFM Analysis

RFM analysis was used to summarize customer behavior using three dimensions. These features provide a compact representation of customer engagement and value.

5.1 Recency

Recency measures the number of days since a customer's most recent purchase. A lower value indicates that the customer has purchased more recently.

5.2 Frequency

Frequency represents the number of purchases/transactions made by a customer. A higher value indicates more frequent purchasing behavior.

5.3 Monetary

Monetary represents the total amount spent by a customer. A higher value indicates greater customer value.

RFM Metric	Mean	Minimum	Maximum
Recency (days)	92.54	1	374
Frequency	4.27	1	209
Monetary	2,054.27	3.75	280,206.02

The RFM table contained 4,338 customers and three behavioral features: Recency, Frequency and Monetary.

6. K-Means Customer Clustering

K-Means clustering was applied to the RFM features to group customers with similar purchasing behavior. Because the RFM variables have different numerical scales, the features were standardized before clustering.

6.1 Selecting the Number of Clusters

Silhouette scores were calculated for K values from 2 through 10. The silhouette score measures how well each observation fits within its assigned cluster compared with other clusters. The highest score occurred at K = 3, with a score of approximately 0.51.

Therefore, three clusters were selected for the final customer segmentation model.

7. Identified Customer Segments

Segment	Customers	Avg Recency	Avg Frequency	Avg Monetary	Total Revenue
High-Value Customers	962	26.90	8.16	3,008.21	2,893,902.43
Regular Customers	2,327	48.97	2.46	731.49	1,702,173.86
At-Risk Customers	1,049	244.86	1.47	427.68	448,637.74

7.1 High-Value Customers

High-Value Customers purchase frequently, have purchased recently and spend considerably more than the other segments. With 962 customers, this group represents about 22% of the customer base but contributes approximately 57% of total revenue.

7.2 Regular Customers

Regular Customers are the largest segment, with 2,327 customers. Their purchasing frequency and monetary value are moderate. This group provides a strong opportunity for upselling, cross-selling and increasing purchase frequency.

7.3 At-Risk Customers

At-Risk Customers have the highest average recency at approximately 245 days and the lowest purchase frequency. Their behavior indicates low recent engagement and makes them suitable targets for customer reactivation campaigns.

8. Power BI Dashboard

The Power BI report was designed as a two-page interactive dashboard. The first page provides an overview of customer segments and their behavioral characteristics. The second page focuses on product preferences and geographic distribution.

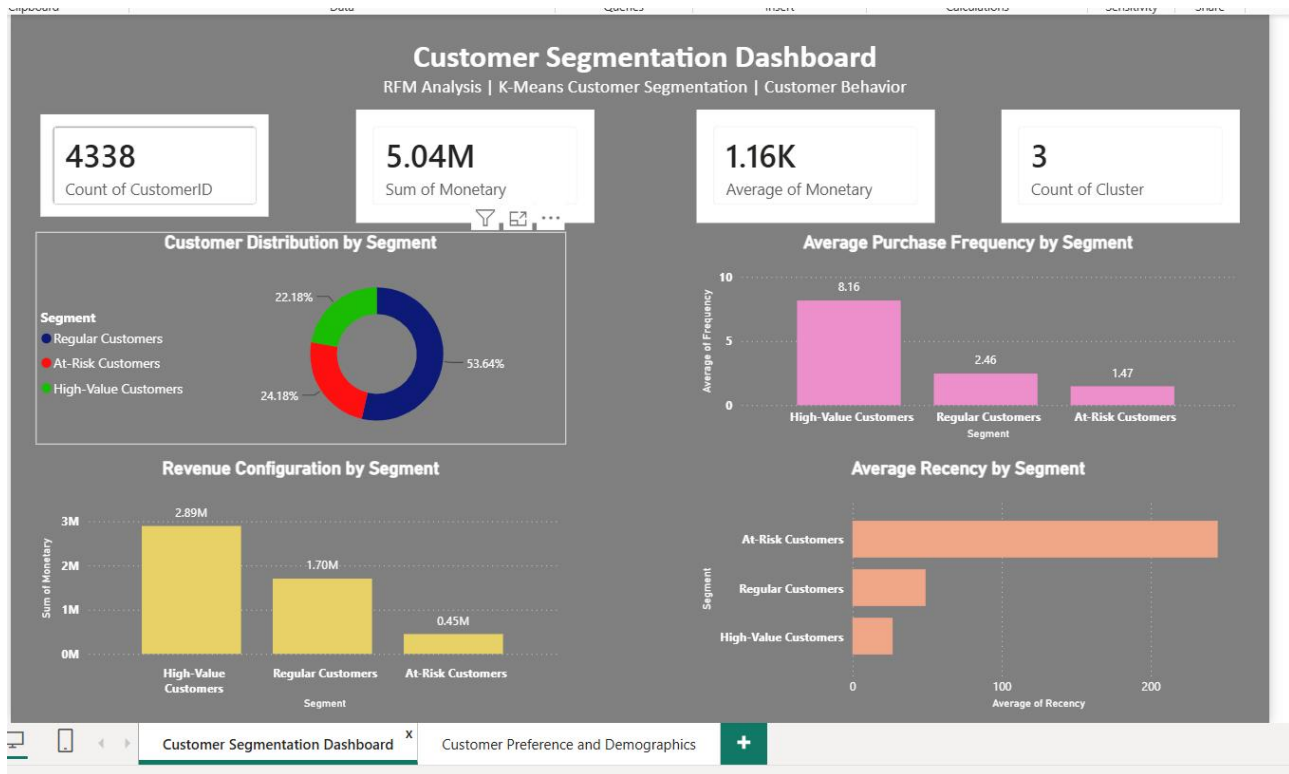


Figure 1. Customer Segmentation Dashboard

The overview dashboard presents total customers, total revenue, average customer spending, number of segments, customer distribution, revenue contribution, average purchase frequency and average recency.

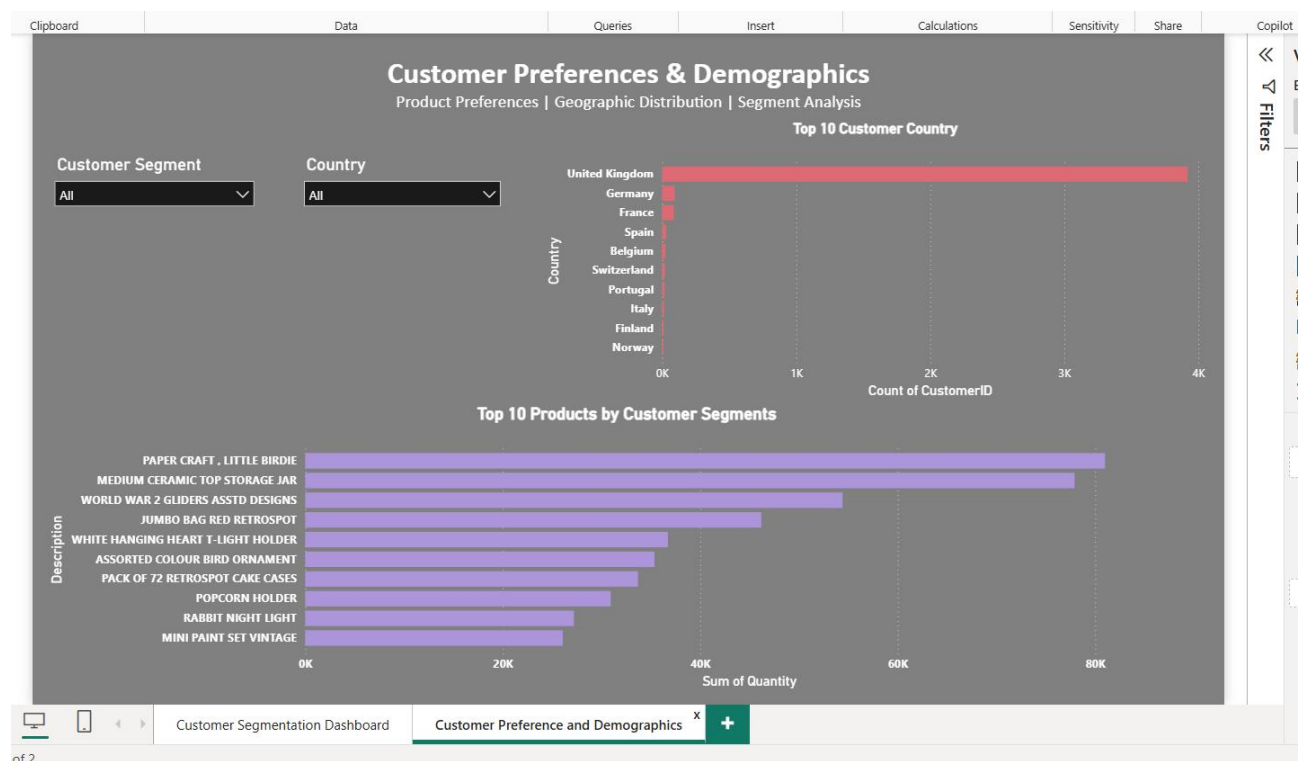


Figure 2. Customer Preferences & Demographics Dashboard

The second dashboard provides interactive Customer Segment and Country filters. It displays the top 10 customer countries and the top 10 products by quantity, allowing purchase preferences to be explored by segment.

9. Key Business Insights

High-value customers drive revenue. High-Value Customers represent about 22% of customers but contribute approximately 57% of total revenue.

High-value customers are more active. Their average purchase frequency is 8.16, substantially higher than Regular Customers (2.46) and At-Risk Customers (1.47).

At-risk customers show weak recent engagement. The At-Risk segment has an average recency of about 245 days, compared with 27 days for High-Value Customers.

Regular customers are the largest segment. With 2,327 customers, Regular Customers represent roughly 54% of the customer base and provide a major opportunity for growth.

Product preferences can support targeted marketing. The product dashboard allows the business to compare high-volume products across customer segments and use those patterns for personalized campaigns.

10. Business Recommendations

10.1 High-Value Customers

- Provide loyalty rewards and exclusive offers.
- Use personalized product recommendations.
- Offer early access to new products or special campaigns.
- Prioritize retention because this segment contributes the largest share of revenue.

10.2 Regular Customers

- Use bundles and cross-selling to increase order value.
- Provide personalized recommendations based on previous purchases.
- Use incentives to increase purchase frequency.
- Move promising regular customers toward the high-value segment.

10.3 At-Risk Customers

- Run targeted win-back and reactivation campaigns.
- Offer limited-time personalized discounts.
- Recommend products related to previous purchases.
- Use email reminders and promotional campaigns to encourage a return purchase.

11. Conclusion

This project successfully segmented 4,338 customers into three meaningful groups using RFM analysis and K-Means clustering. The resulting segments—High-Value, Regular and At-Risk Customers—show clear differences in purchase frequency, recency and monetary value.

The analysis demonstrates that a relatively small group of high-value customers contributes a large share of revenue, while at-risk customers require targeted reactivation efforts. The Power BI dashboard converts these analytical results into an interactive business tool for exploring customer segments, product preferences and geographic characteristics.

Overall, the project demonstrates a complete customer analytics workflow: data cleaning, feature engineering, RFM analysis, clustering, visualization and business recommendation.

12. References

- Kaggle – Online Retail Dataset.
- Pandas – Python data analysis library.
- Scikit-learn – Machine learning library used for K-Means clustering and silhouette analysis.
- Microsoft Power BI – Business intelligence and visualization platform.